APPLICATION DEVELOPMENT WITH VIEWS

ISE305 – DATABASE DESIGN & PRACTICE

SOME SLIDE MATERIAL FROM WATSON

Reading & Reference

- Form design
  - http://www.youtube.com/watch?v=Rj0ZBLlbsRc
  - http://office.microsoft.com/en-us/access-help/create-an-access-form-HA102749786.aspx

Outline

- Background
- Creating views and using views
- Processing queries that reference views
- Updatable views
- Data requirements for hierarchical forms

What is a View?

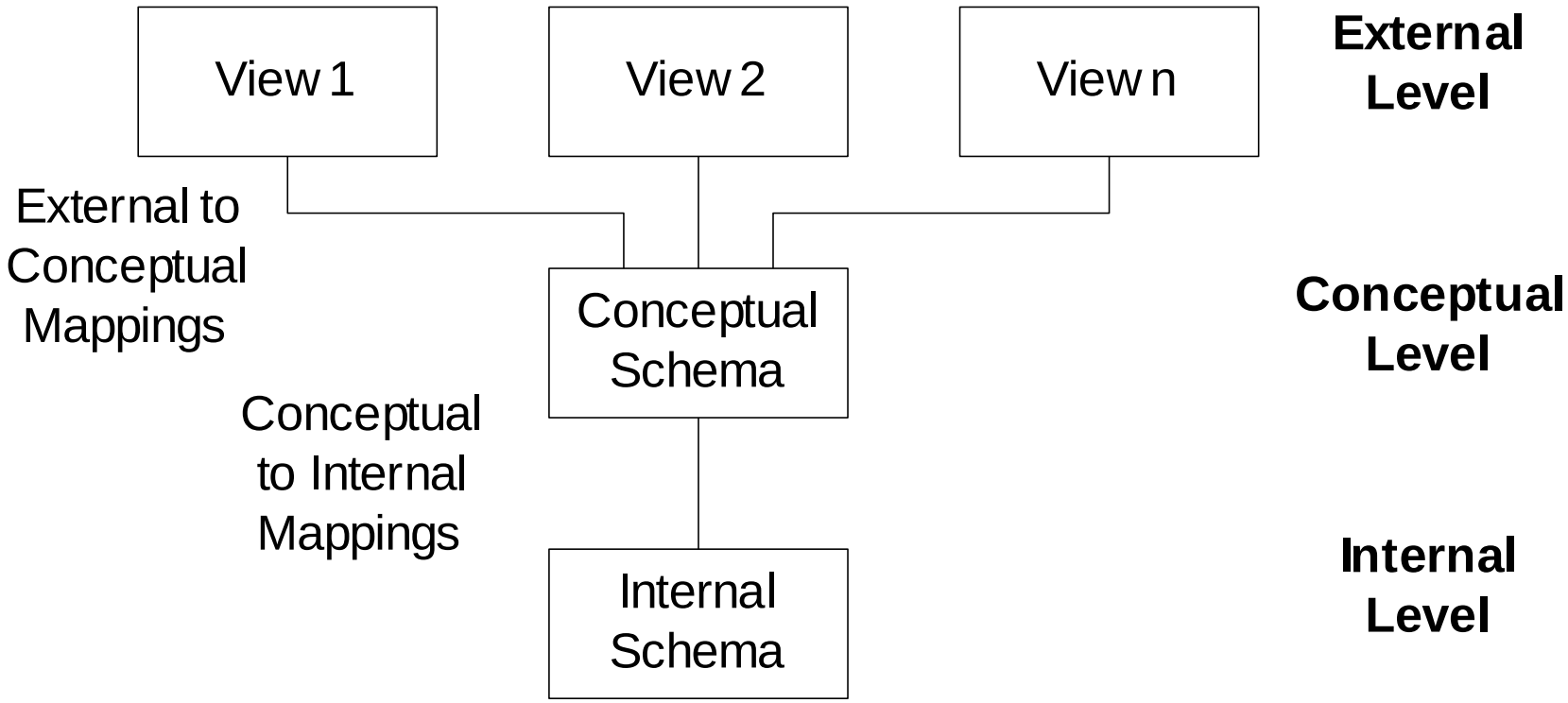
- Derived table (virtual table)
- Behaves like a base table, but no physical table exists
- Stored query

View Advantages

- Reduce impact of database changes
- Simplify database usage
- Prevents users from accessing data they are not permitted to see
- Customizes the appearance of the DB

Might be a performance penalty on complex views

Three Schema Architecture



View Definition Example

- Create a view - offerings taught by faculty in the MS department.

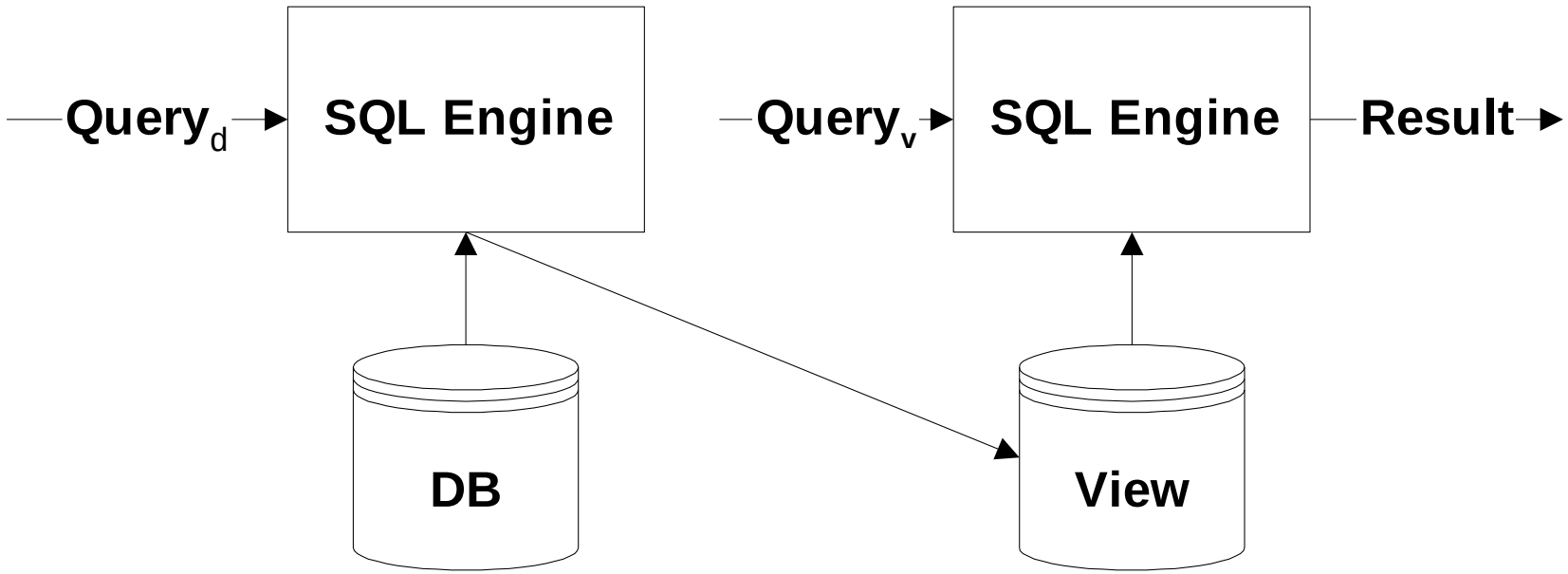
Name of view

Column names are the same

```
CREATE VIEW MS_View AS
 SELECT OfferNo, Course.CourseNo, CrsUnits,
        OffTerm, OffYear, Offering.FacSSN,
        FacFirstName, FacLastName, OffTime,
        OffDays
  FROM Faculty, Course, Offering
  WHERE FacDept = 'MS'
    AND Faculty.FacSSN = Offering.FacSSN
    AND Offering.CourseNo = Course.CourseNo;
```

We will use the CREATE VIEW SQL statement once we begin to use MySQL

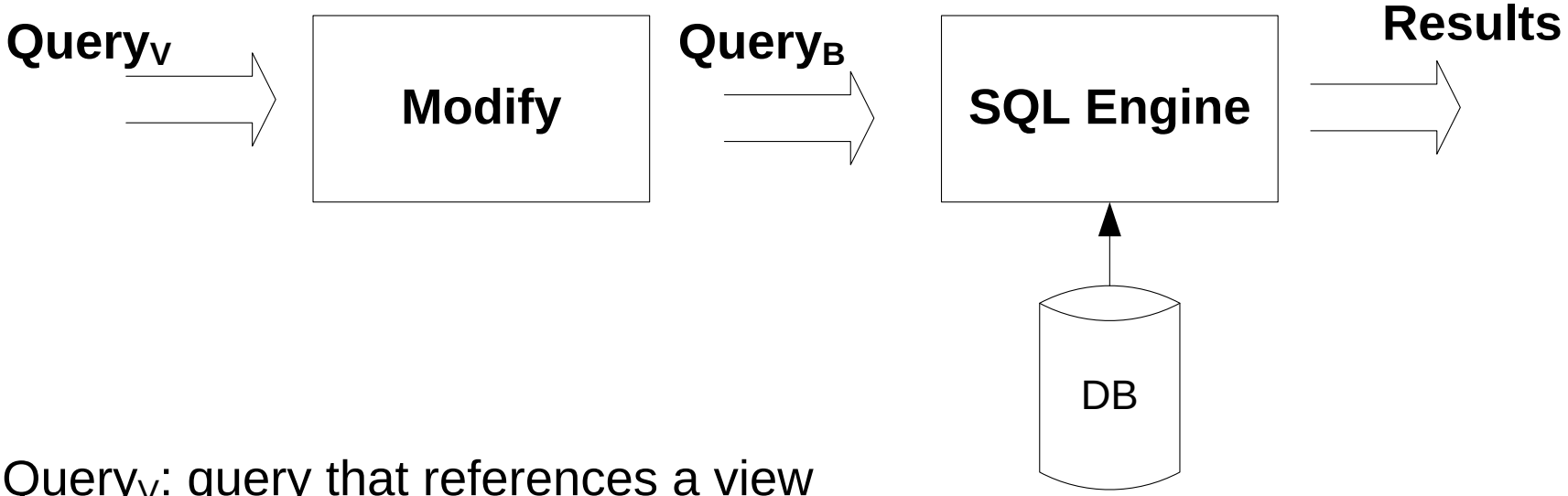
View Materialization



Query$_d$: Query that defines a view

Query$_v$: Query that references a view

View Modification



Query$_V$: query that references a view

Query$_B$: modification of Query$_V$ such that references to the view are replaced by references to base tables.

Single Table Updatable Views

- A view can be read-only or updatable
- Support modification statements
- Additional rules for single table updatable views

Hierarchical Forms

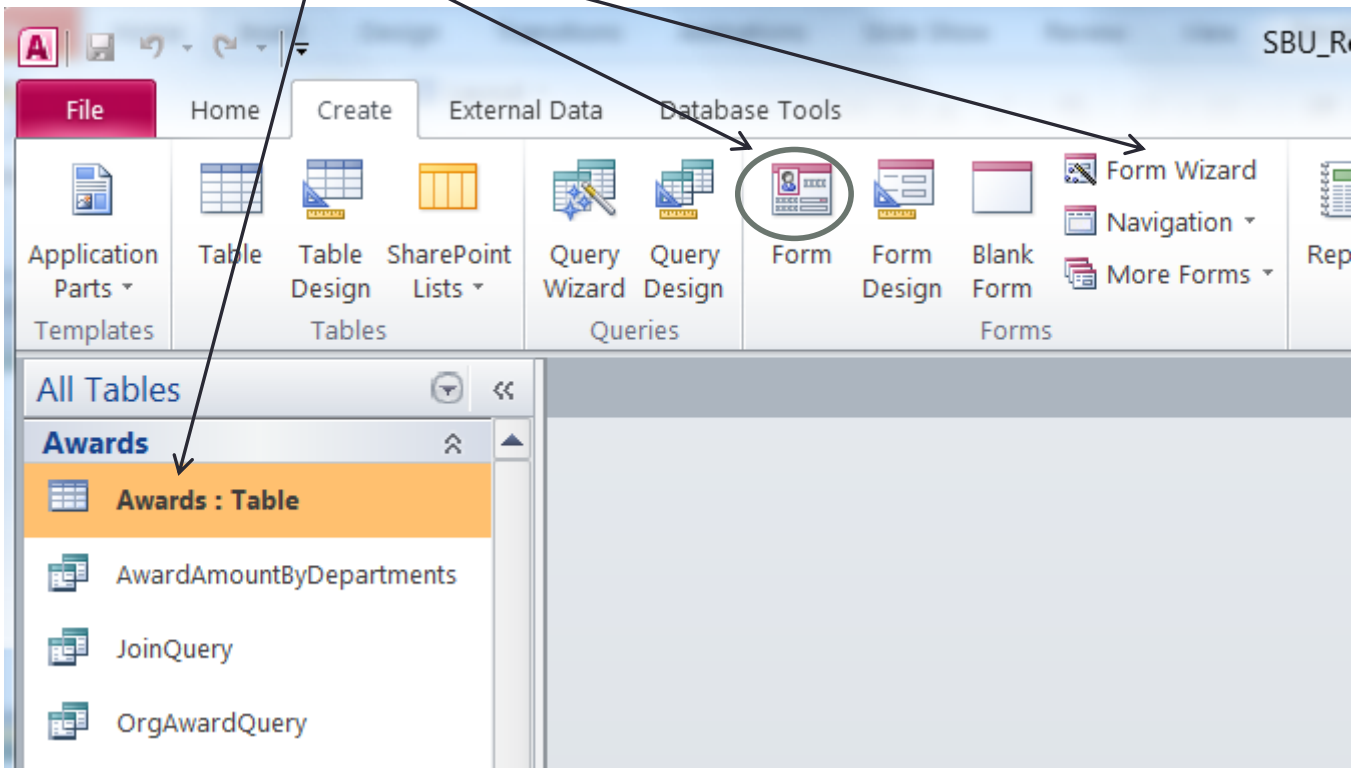
- Formatted window for data entry and display
- Contains
  - Main form
  - Sub-form
- Provides attractive interface for a 1-M relationship

A form is a screen that allows you to enter, change, and view data in a DB

The term “hierarchal form” or “sub-form” is used to describe a hierarchy (tree) with 1-M relationships

Create a Form

- Select table and the Create Form



Forms

- Forms are made up of controls, (e.g., text boxes, buttons, etc.), grouped in a way that makes them easy to use
- The controls in the form are usually bound, or linked, to the tables or queries in your database
- Other uses
  - you can create a form that asks for input
  - generate a custom report based on that input

Types of Forms

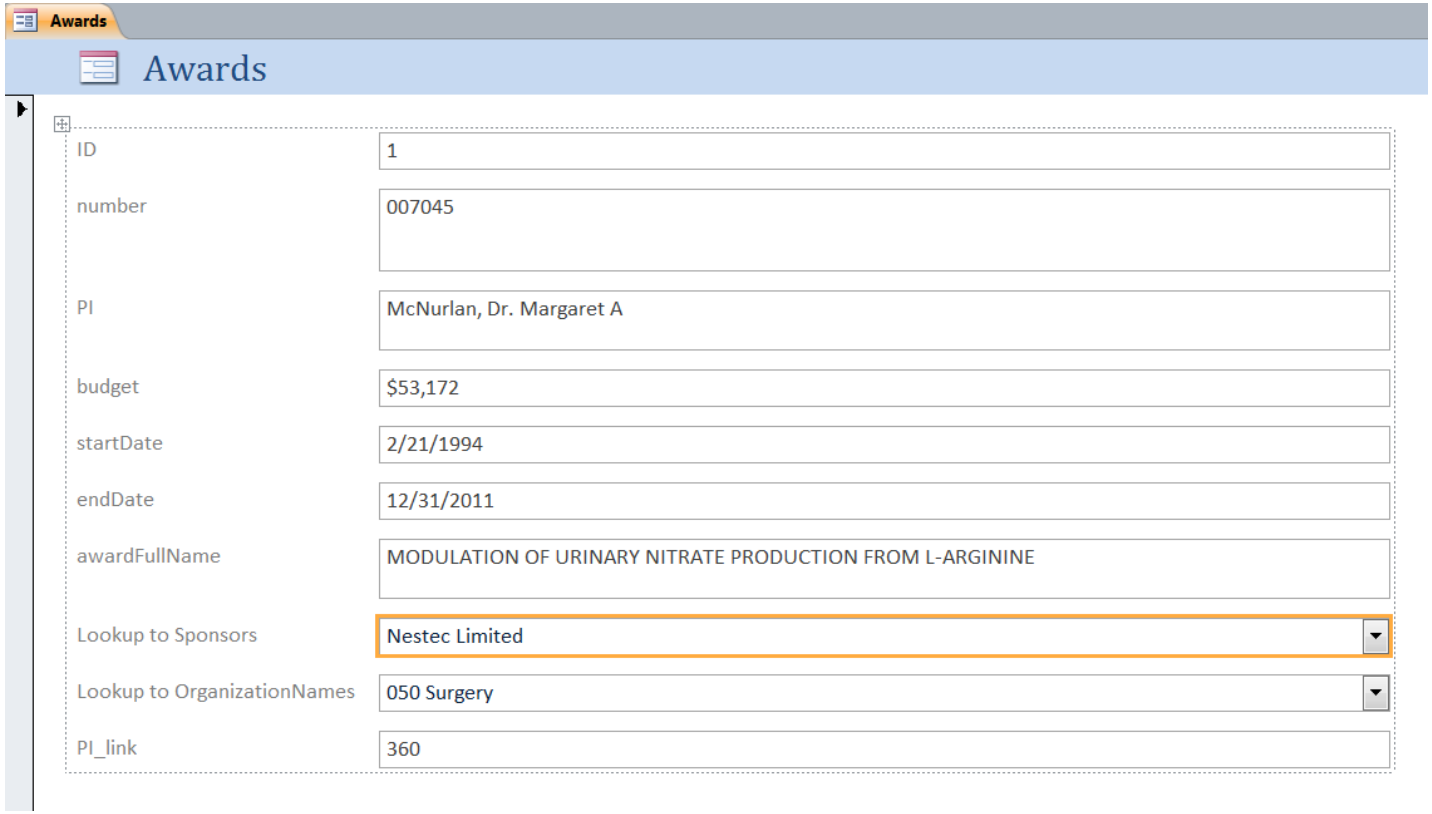
- Detail – single data source (table or query)
- Split form – Single source detail form and data sheet (changes only one record at a time)
- Subforms (parent-child forms) – for 1-m relationships
- Multiple item forms - like a data sheet, but more control
- Navigation form – uses tabs and/or side menus

Form Design

- Access provides tools to easily design forms using the following steps:
  - Select a data source
  - Select a form type
- Once you have created an initial form, you can rearrange or delete controls

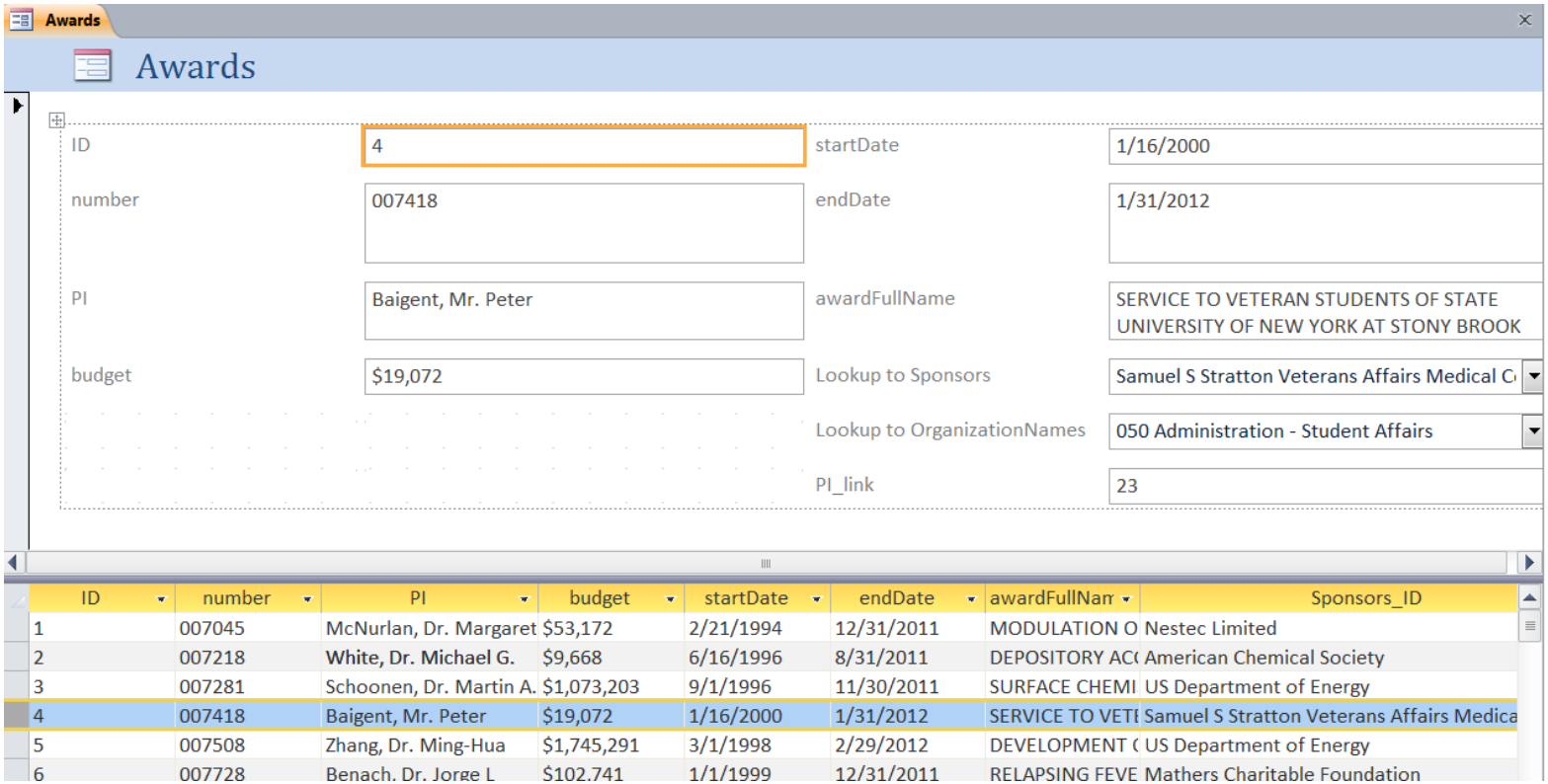
Example – Detail Form

- In the SBU_Awards DB, clicking Awards table and Form design obtains



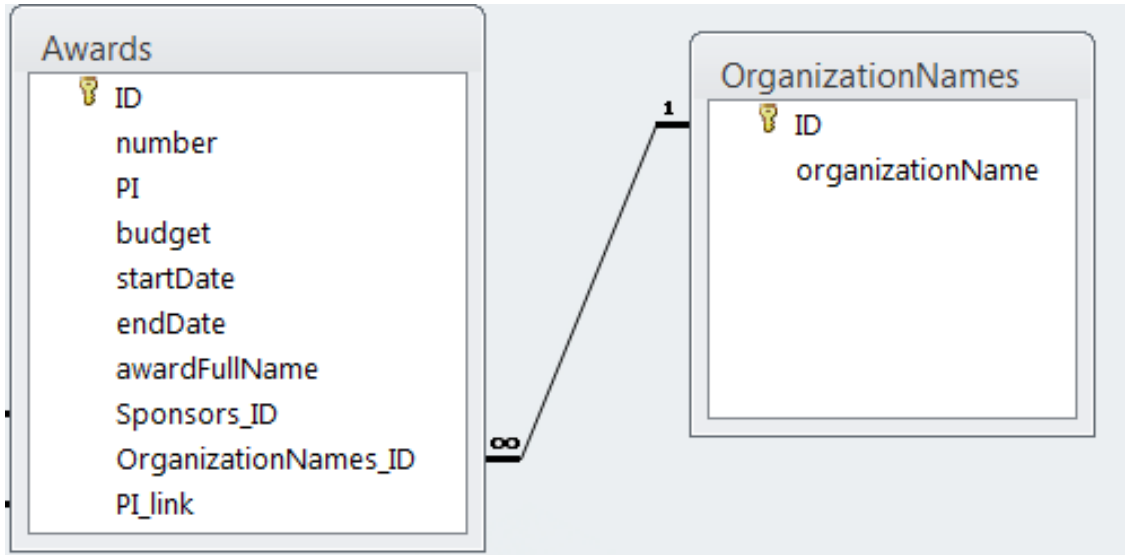
Split Form

- In the SBU_Awards DB, clicking Awards table and Split Form obtains



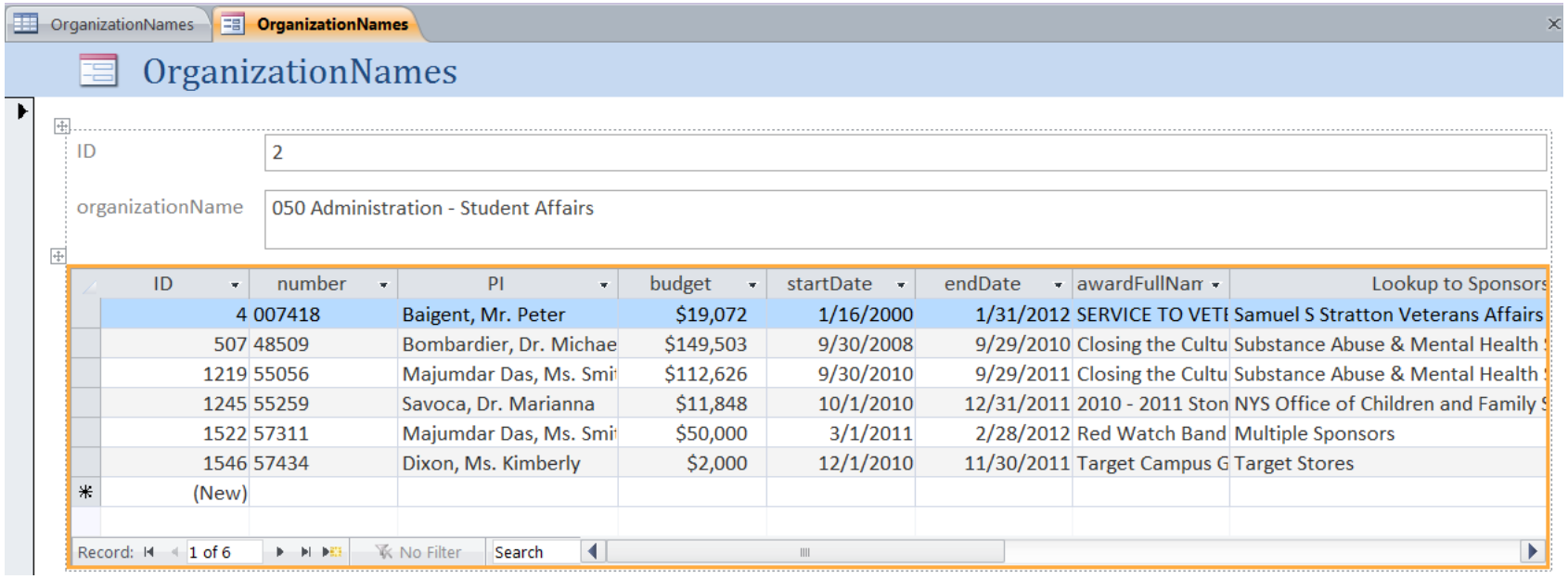
Subform

- You can create a subform easily if the parent table has only one 1-m relationship



Subform Example

- Selecting OrganizationNames table and clicking Create Form obtains



ID	number	PI	budget	startDate	endDate	awardFullNam	Lookup to Sponsors
4	007418	Baigent, Mr. Peter	$19,072	1/16/2000	1/31/2012	SERVICE TO VETI	Samuel S Stratton Veterans Affairs
507	48509	Bombardier, Dr. Michae	$149,503	9/30/2008	9/29/2010	Closing the Cultu	Substance Abuse & Mental Health
1219	55056	Majumdar Das, Ms. Smi	$112,626	9/30/2010	9/29/2011	Closing the Cultu	Substance Abuse & Mental Health
1245	55259	Savoca, Dr. Marianna	$11,848	10/1/2010	12/31/2011	2010 - 2011 Ston	NYS Office of Children and Family
1522	57311	Majumdar Das, Ms. Smi	$50,000	3/1/2011	2/28/2012	Red Watch Band	Multiple Sponsors
1546	57434	Dixon, Ms. Kimberly	$2,000	12/1/2010	11/30/2011	Target Campus G	Target Stores
(New)							

Create Subform Manually

- To manually create a subform (when the parent table is the 1 side of more than one 1-m relationship)
  - Create a detail form for the table on the 1 side
  - Create a datasheet form for the table on the many side
  - Save the forms
  - Drag the datasheet form to the detail form
  - Click the build button in the Link Master Fields property of the child form
  - Link the two tables

Details in reference

Change Layout of Form

- You can:
  - Change the layout of your form
  - Add form elements
  - Delete form elements
- Form components are arranged in a grid using the Layout view of the form
- Most changes involve drag and drop actions on your form layout
- Details in video on page 6 of reference

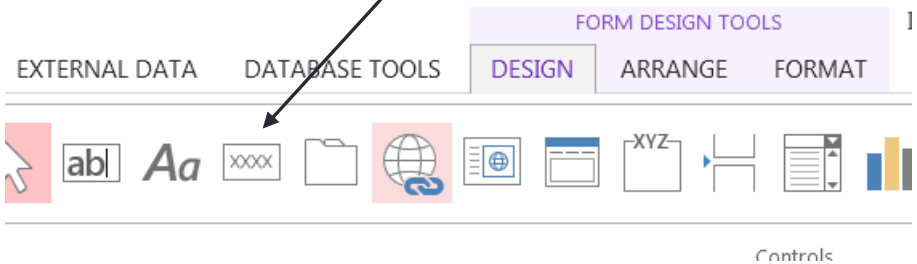
You can also use the design view to change the layout of a form

Sizing Your Form

- Design your forms to fit on your users' screens
- Example – a 1024×768 screen
  - A form to fit this size should be about 9.75 inches wide
  - the sum of the heights of the sections should be about 5.6 inches to allow space for the Ribbon, status bar, and Windows taskbar.

Add Navigation

- You can add controls to navigate through records in your DB
- In Form Design Tools, click on Button
- A pop-up box enables you to select the type of navigation associated with the button (e.g., go to next record)



Analysis of Data Requirements

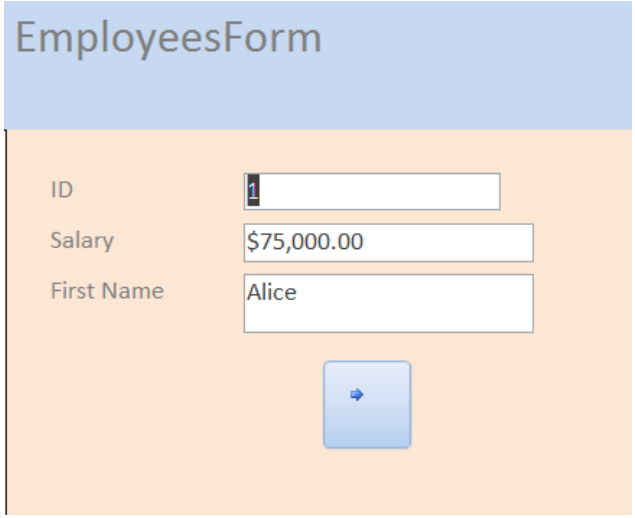
- Identify the 1-M relationship
- Identify the linking fields
- Determine other tables in the main form and the subform
- Determine updatable tables
- Write queries for the main form and subform

Think about the forms you will need in your project DB

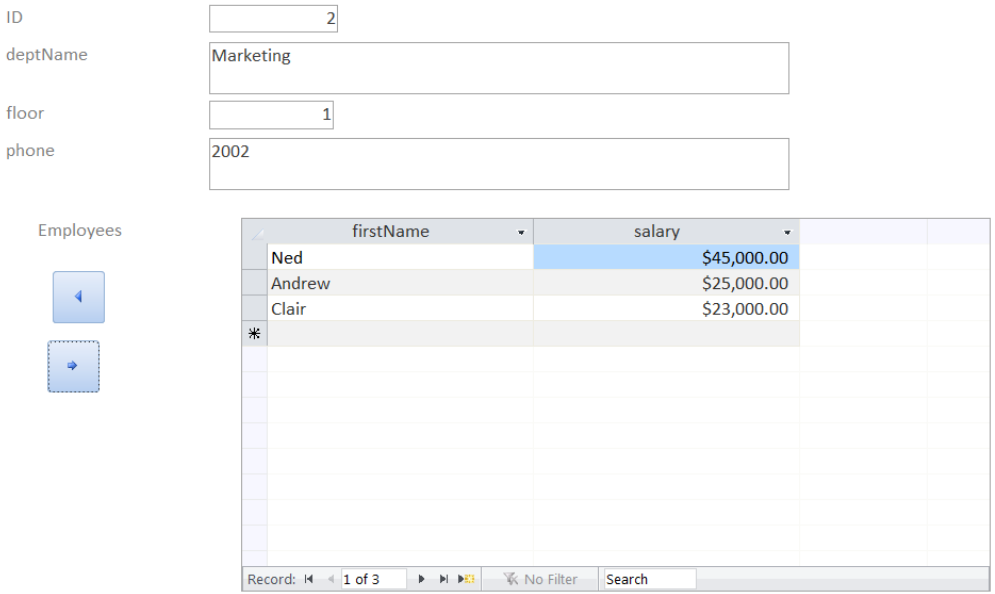
Are We on Track?

- For the Employees DB, create:
  - A single table form (columnar) for the Employees table (use the wizard and select the ID, first name, and salary fields)
  - In the Form View, change the name and salary of one of the employees
  - Using Form Layout tools, change the appearance of the form
  - Using Form Design View, change the appearance (e.g., titles, width of fields, etc.)
  - In Design View add a navigation button (for the next record)

In case you do not have your tables, download Excel files from: www.cs.stonybrook.edu/~ise305/Employees.xlsx and www.cs.stonybrook.edu/~ise305/Departments.xlsx



Choose your own appearance. The above is only an example.

Are We on Track?

- For the Employees, create a Department form that has an Employee sub-form
- Use the Form Wizard
- Create an interesting design (not example design)
- Include a previous button